

Coding & Solution

Date	30-June-2026
Team ID	
Project Name	A comprehensive measure of well being
Team Members	Chikali Sarika, Anusha Gollapalli, Karishma Sharon Kaparapu, Ambika Chitikela, Parneeth Devanaboina
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of the implemented HDI Predictor code and overall solution, demonstrating working functionality, clean code practices, and alignment with the defined project requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	<a href="https://github.com/<your-team-username>/HDI-Predictor">https://github.com/<your-team-username>/HDI-Predictor (replace with your actual GitHub/GitLab repository link)
Programming Language(s)	Python
Framework(s) Used	Flask (web app), scikit-learn (ML model), Pandas & NumPy (data processing), Matplotlib & Seaborn (visualization)
Key Features Implemented	- Data preprocessing pipeline for UNDP/Kaggle HDI indicators (life expectancy, schooling years, GNI per capita). - Trained scikit-learn classification model predicting HDI tier (Very High/High/Medium/Low). - Flask web form for entering country indicators and receiving instant predictions. - Result page displaying the predicted HDI tier with a short explanation. - Matplotlib/Seaborn charts showing indicator contributions to the prediction. - Coverage of all three project scenarios: high-performing, mid-range, and low-development countries.
Pending / Incomplete Features	- Cloud deployment (currently runs locally / on a development server). - User authentication for saving prediction history. - Automated retraining pipeline when new country data is published.
Setup / Run Instructions	1. Clone the repository and open it in Anaconda/PyCharm. 2. Create an environment and install dependencies: <code>pip install flask scikit-learn pandas numpy matplotlib seaborn</code> 3. Run the training script once to generate the model file: <code>python train_model.py</code> 4. Start the web app: <code>python app.py</code>

	5. Open http://127.0.0.1:5000 in a browser and enter indicator values to get an HDI prediction.
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Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Partial
4	Error handling is implemented for critical operations	Partial
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The current implementation covers all three project scenarios end-to-end. Remaining work before final submission is cloud deployment and expanding inline documentation and error handling around file I/O operations.